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ACADEMIC VAULT

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FEG 101: Engineering Mathematics I - Functions, Calculus and Vectors

The mathematical toolkit that every first-year engineering course assumes you already have

FACULTY PRIMER

Faculty Primer

Faculty of Engineering
Engineering (Common 100L Courses) | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

Engineering students fail FEG 101 for two reasons: weak algebra and rushing calculus without understanding limits. This pack rebuilds the algebra foundations, explains differentiation and integration through physical meaning, and drills vector operations with engineering-flavoured worked examples.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Manipulate algebraic expressions, indices, logarithms and partial fractions without calculator dependence.
- Evaluate limits and differentiate standard functions, including product, quotient and chain rules.
- Integrate by substitution and by parts, and apply definite integrals to area and work problems.
- Add, resolve and take dot and cross products of vectors in two and three dimensions.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Repair the algebra first

Most 'calculus problems' are actually algebra problems wearing a disguise. Fix these and the rest of the course becomes manageable.

- Indices: a^m x a^n = a^(m+n); a^(m/n) is the n-th root of a^m.
- Logarithms: log(ab) = log a + log b; log(a^n) = n log a. Know the change-of-base rule.
- Factorise before you substitute. A common factor often cancels an apparent difficulty.
- Partial fractions: split a rational expression into simpler denominators before integrating.
- Quadratic formula and completing the square reappear in optimisation and in dynamics.

> **No-calculator drill**
> Spend 15 minutes weekly evaluating simple expressions by hand. Engineering exams reward speed and accuracy, and both come from fluency, not from the calculator.

Limits and differentiation

The derivative is an instantaneous rate of change. Understand that sentence and the rules become tools rather than magic.

The derivative of f at x is the limit of the change in f divided by the change in x as the change approaches zero. That is why velocity is the derivative of displacement and current is the derivative of charge.

Learn the standard derivatives, then the combination rules: product, quotient and chain. The chain rule is the one students under-practise, and it appears in almost every engineering application.

- d/dx of x^n is n x^(n-1); of sin x is cos x; of cos x is -sin x; of e^x is e^x; of ln x is 1/x.
- Product rule: (uv)' = u'v + uv'.
- Quotient rule: (u/v)' = (u'v - uv') / v^2.
- Chain rule: differentiate the outside, then multiply by the derivative of the inside.
- Use the first derivative for increasing/decreasing behaviour and the second for concavity.

Integration and vectors

Integration reverses differentiation and accumulates quantity. Vectors describe quantities with both size and direction.

- A definite integral gives signed area. Always sketch the curve to check whether your answer is plausible.
- Substitution works when you can see the derivative of an inner function in the integrand.
- Integration by parts follows the product rule; choose u so that differentiating it simplifies the expression.
- Resolve a vector into components: Fx = F cos(theta), Fy = F sin(theta).
- Dot product gives work and the angle between vectors; cross product gives torque and a perpendicular vector.

> **Units are a check**
> If you are integrating a velocity in m/s over time in seconds, the answer must be in metres. If your units are wrong, your method is wrong.

Functions, limits and continuity

Engineering mathematics models physical systems. A function describes how one quantity depends on another; its domain and range describe what is physically meaningful. Choosing the right function - linear, exponential, sinusoidal - is a modelling decision before it is a mathematical one.